

their spin—a quantum-mechanical property that forms the basis of magnetic data storage. Engineers hope to exploit the spin-motion connection to make superfast hard drives.

Topological insulators open the door to tailoring topological electronic properties by stacking different thin sheets, or 2D materials. These exotic 2D materials could be used as a platform for energy-efficient computing (spintronics) and to solve today's intractable challenges with quantum computing.

Candidate materials for topological insulators

Tungsten ditelluride. Like graphene, the semi-metal tungsten ditelluride (WTe_2) can be prepared in a single monolayer. Tellurium atoms sandwich the transition metal tungsten in each layer. These sandwiched transition metal materials are important for future electronics and photonics. Scientists have predicted that WTe_2 in monolayer form has the exotic electronic properties of topological

insulators. However, the surface of WTe_2 oxidises in air, destroying the electronic properties.

Now, researchers have made devices from WTe_2 down to a single layer thick, which are air-stable and have good electrical contacts. Surprisingly, they found that in the case of a single layer, the sheet became insulating at liquid nitrogen temperatures when no gate voltage was applied. For large-enough positive or negative contact voltages, the electrical current switched on, as in a transistor.

Most remarkably though, the insulating state was not fully insulating—it still conducted at the edges. Multi-contact electrical transport measurements and scanning probe techniques verified this conductivity at the edge. This edge conduction is just what is expected for a 2D topological insulator. It exhibits another topological signature in its electronic spectrum—electron states at all energies, without the gap normally found in semiconductors and insulators. The special edge conductivity is sensitive

to a magnetic field pointing parallel to the layer.

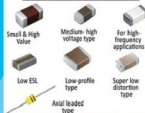
These observations are consistent with the edge conduction channels behaving like quantum mechanical wires in which the electron magnetic direction, known as spin, is locked to the direction of electron travel. Therefore the edge of the material can be used to filter electrons by spin direction.

The only previous 2D topological insulator candidates were quantum wells in sophisticated semiconductor heterostructures. Likewise, this 2D material can readily be combined with layers of superconducting and magnetic materials. It may thus provide the basis of a platform for spin-based or topologically protected quantum computing.

Transition metal oxides (TMOs). These are a class of materials that share a common crystal structure known as perovskite. They consist of oxygen and a transition metal element such as copper, iron, manganese, cobalt or titanium. The

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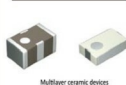
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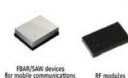
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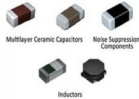
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